

Xiuqi Bai

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EDUCATION

University of Waterloo

Bachelor of Software Engineering, Honours

Sep 2024 – Apr 2029

4.0/4.0 GPA

EXPERIENCE

SWE Intern

Shopify Inc.

Jan 2026 – Apr 2026

Toronto, Canada

- Architected a **backward-compatible testing framework** across monolithic codebases, decoupling test harnesses to eliminate **100%** of asynchronous race conditions and non-deterministic flaky failures.
- **Cut CI test execution runtime by 50%** by implementing graph-based selective test running and reducing runtime mounting overhead, saving hundreds of developer engineering hours weekly.
- Developed an **automated AI coding-agent workflow** to synthesize deterministic regression test suites and interface mocks, enforcing strict architectural contracts across core services.
- Delivered **10+** mission-critical merchant-facing services and dashboard workflows, enhancing data visibility and operational reliability for millions of active businesses.

MLE Intern

Huawei Technologies Canada Co., Ltd.

May 2025 – Aug 2025

Ottawa, Canada

- Engineered **high-performance CNN architectures** in **PyTorch** for real-time signal calibration, boosting signal-to-noise ratio (SNR) from **20 dB to 40 dB** and winning selection over 5 competing architectures for **global production deployment**.
- Built an **end-to-end transceiver simulation pipeline** using **NumPy** and **SciPy**, generating synthetic high-fidelity training datasets to accurately model optical physical layers at scale.
- Reduced **model training runtime by >50%** by implementing dynamic learning-rate scheduling via LR range tests and optimizing custom loss convergence.
- Developed **automated diagnostics in Matplotlib** to inspect convolutional weight tensors, isolating channel imbalances across time-frequency domains to ensure production numerical stability.

PROJECTS

Embedded Voice Assistant |

Mar 2025 – May 2025

- Engineered a standalone edge voice assistant on an **ESP32** in **C++**, implementing non-blocking event loops and finite-state machines to manage concurrent audio processing, Wi-Fi telemetry, and peripheral control.
- Configured low-latency **I2S digital audio pipelines** with Direct Memory Access buffers, fine-tuning sample rates and ring-buffer thresholds to capture and playback synchronized high-fidelity audio.
- Built a local **Python/FastAPI inference backend** hosting a quantized open-source LLM, orchestrating an end-to-end pipeline spanning speech-to-text, context generation, and text-to-speech synthesis.
- Implemented bi-directional **WebSocket streaming** between the ESP32 and host server, serializing chunked raw audio frames with automated reconnection logic to achieve sub-second end-to-end voice latency.

Moodii |

Apr 2024 – Aug 2024

- Developed an AI-powered Android app in **Kotlin** with Jetpack Compose, enabling users to log moods and track emotion trends via voice recordings.
- Designed and implemented a **Spring Boot** backend with **15+** RESTful APIs, enabling secure user authentication, efficient data storage, and advanced emotion trend analytics.
- Integrated a **CNN+LSTM** Speech Emotion Recognition model through **Flask**, achieving **81%** accuracy.
- Orchestrated the containerization of backend and ML services using **Docker**, deploying to **Azure** with automated CI/CD pipelines to ensure rapid, reliable deployments and horizontal scalability.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Kotlin, SQL, Go, HTML & CSS, Typescript & JavaScript

Frameworks and Libraries: PyTorch, TensorFlow, Spring Boot, Node.js, React.js, Django, NumPy

Developer Tools: AWS, Azure, Docker, MongoDB, MySQL, Git, GitHub, Linux, Postman, Docker